Chapter 4 Question Bank

One hundred questions on probability and counting rules, every one labelled with the section it comes from, every one solved in full.

100 questions | **Part A** 25 concept MCQs | **Part B** 53 section drills | **Part C** 10 exam-style | **Part D** 12 midterm simulation | **CLO** 4-A, 1

HOW TO USE THIS BANK

1. Work **questions only** — the solutions live in a separate half of the document, far below. Do not scroll down until you have written an answer.
2. Every question carries a **section tag** (§4-1 §4-2 §4-3 §4-4 / §4-5). If you miss a question, the tag tells you exactly which lesson to re-read in the summary document.
3. Use the → **solution** link on each card to jump to its worked answer, and the ← **back** link to return.
4. Part D is a **timed 60-minute midterm simulation** covering only Chapter 4. Sit it once a week before the exam, closed-book.
5. Record your score in the tracker at the end and re-attempt every question you missed after three days — not the same day.

طريقة الاستخدام

حلّ **الأسئلة أولًا** دون النظر إلى الحلول — الحلول موجودة في النصف الثاني من الملف. كل سؤال عليه **وسم القسم** الذي جاء منه (4-1 / 4-2 / 4-3 / 4-4 و 4-5)، فإذا أخطأت في سؤال فالوسم يدلّك مباشرةً على الدرس الذي يجب مراجعته في ملف الملخص. الجزء **D** امتحان محاكٍ للميدتيرم مدته **60 دقيقة** — اجلس له مغلق الكتاب قبل أسبوع من الامتحان، وأعد حلّ ما أخطأت فيه بعد **ثلاثة أيام** وليس في اليوم نفسه.

JUMP TO

A	Concept & definition MCQs (A1–A25)	B·1	§4-1 drills (B1–B13)
B·2	§4-2 drills (B14–B23)	B·3	§4-3 drills (B24–B37)
B·4	§4-4 & §4-5 drills (B38–B53)	C	Exam-style structured (C1–C10)
D	Midterm simulation (D1–D12)	SOL	Full worked solutions
TR	Score tracker		

PART A

Concept and definition questions

أسئلة المفاهيم والتعاريف — 25 سؤال اختيار من متعدد

These are CLO1 marks: fast, verbatim, and routinely worth the difference between A– and A. Aim for 25/25 in under eight minutes.

A1 §4-1 → sol

The result of a single trial of a probability experiment is called:

(a) a sample space (b) an outcome (c) an event
(d) an experiment

A2 §4-1 → sol

The set of all possible outcomes of a probability experiment is called:

(a) an event (b) an outcome (c) the sample space
(d) a simple event

A3 §4-1 → sol

An event consisting of exactly one outcome is called:

(a) a compound event (b) a simple event (c) an impossible event
(d) a certain event

A4 §4-1 → sol

If $P(E) = 1$, then the event E is:

(a) impossible (b) unlikely (c) certain

(d) equally likely

A5 §4-1 → sol

Which of the following **cannot** be a probability?

(a) 0 (b) 0.0001 (c) 5/4

(d) 1

A6 §4-1 → sol

A frequency distribution built from a survey of 200 people is used to compute a probability. This probability is:

(a) classical (b) empirical (c) subjective

(d) conditional

A7 §4-1 → sol

"A seismologist states there is an 80% probability that an earthquake will occur in a certain area." This is an example of:

(a) classical probability (b) empirical probability (c) subjective probability

(d) complementary probability

A8 §4-1 → sol

When two dice are rolled, $n(S)$ equals:

(a) 12 (b) 36 (c) 6

(d) 21

A9 §4-1 → sol

For the gender of the children in a family with three children, $n(S)$ equals:

(a) 3 (b) 6 (c) 8

(d) 9

A10 §4-1 → sol

Consider the values 2, 3, 4, 5, 6. The phrase "**at most 4**" refers to which values?

(a) 2, 3 (b) 2, 3, 4 (c) 3, 4

(d) 5, 6

A11 §4-1 → sol

Consider the values 2, 3, 4, 5, 6. The phrase "**more than 4**" refers to which values?

(a) 4, 5, 6 (b) 5, 6 (c) 2, 3

(d) 2, 3, 4

A12 §4-2 → sol

Two events that have no outcomes in common are called:

(a) independent (b) dependent (c) mutually exclusive

(d) complementary

A13 §4-2 → sol

If A and B are mutually exclusive, then $P(A \cap B)$ equals:

(a) $P(A) \cdot P(B)$ (b) 0 (c) 1

(d) $P(A) + P(B)$

A14 §4-2 → sol

The formula $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ is:

(a) valid for any two events (b) valid only when A, B are mutually exclusive (c) valid only when A, B are independent

(d) never valid

A15 §4-2 → sol

In probability, the statement "A **or** B" means:

(a) A only (b) B only (c) A, or B, or both

(d) exactly one of A and B

A16 §4-3 → sol

Two events A and B are **independent** if:

(a) $A \cap B = \varnothing$ (b) $P(B \mid A) = P(B)$ (c) $P(A) + P(B) = 1$

(d) they cannot occur together

A17 §4-3 → sol

Selecting two items from a box "**without replacement**" makes the two selections:

(a) independent (b) dependent (c) mutually exclusive

(d) complementary

A18 §4-3 → sol

The conditional probability $P(B \mid A)$ equals:

(a) $P(A \cap B) / P(B)$ (b) $P(A \cap B) / P(A)$ (c) $P(A) \cdot P(B)$

(d) $P(A \cup B) / P(A)$

A19 §4-3 → sol

When **two** separate coins are tossed, the result of the first toss and the result of the second toss are:

(a) mutually exclusive (b) independent (c) complementary

(d) dependent

A20 §4-3 → sol

The probability of getting **at least one** success equals:

(a) $1 - P(\text{all})$ (b) $1 - P(\text{none})$ (c) $P(\text{none})$

(d) $1 + P(\text{none})$

A21 §4-4 → sol

The value of $0!$ is:

(a) 0 (b) 1 (c) undefined

(d) infinite

A22 §4-4 → sol

When the **order** of selection matters, you should use:

(a) a combination (b) a permutation (c) the complement rule

(d) the addition rule

A23 §4-4 → sol

The number of ways to form a committee of 4 people from 10, where no member holds a special title, is:

(a) ${}_{10}P_4$ (b) ${}_{10}C_4$ (c) $4!$

(d) $10!$

A24 §4-4 → sol

The relationship between permutations and combinations is:

(a) ${}_nC_r = {}_nP_r \times r!$ (b) ${}_nC_r = {}_nP_r \div r!$ (c) ${}_nC_r = {}_nP_r + r!$

(d) ${}_nC_r = {}_nP_r - r!$

A25 §4-5 → sol

In `P(E) = n(E)/n(S)`, if `n(S)` has been counted using combinations, then `n(E)` must be counted using:

(a) permutations (b) combinations (c) factorials

(d) either one

PART B·1

§4-1 drills — sample spaces, classical and empirical probability

تدريبات القسم 4-1: فضاء العينة، الاحتمال الكلاسيكي والتجريبي، المتمّمة

Write the sample space in set notation for every question where it is small enough. The lecturer states explicitly that set notation is required.

B1 §4-1 → sol

A single die is rolled. Write the sample space S and find the probability of getting an even number.

B2 §4-1 → sol

A single die is rolled. Find the probability of getting a number greater than 4.

B3 §4-1 → sol

Two dice are rolled. Find the probability that the sum of the two faces is 7. List the favourable outcomes.

B4 §4-1 → sol

Two dice are rolled. Find the probability that the sum is **at least 10**.

B5 §4-1 → sol

Two dice are rolled. Find the probability of getting doubles (both dice showing the same number).

B6 §4-1 → sol

A family has three children. Find the probability that **at least two** of them are boys.

B7 §4-1 → sol

A family has three children. Find the probability that **none** of them is a girl.

B8 §4-1 → sol

One card is drawn at random from a standard deck of 52 playing cards (4 suits of 13 cards; the suits are hearts, diamonds, clubs and spades; each suit contains one king). Find: (a) P(the card is a heart); (b) P(the card is a king).

B9 §4-1 → sol

A survey of 80 people recorded their preferred drink: tea 34, coffee 28, juice 18. For a person selected at random, find: (a) P(tea); (b) P(not coffee); (c) P(tea or juice).

B10 §4-1 → sol

Using the drink survey of B9, find the probability that a randomly selected person prefers **neither tea nor coffee**.

B11 §4-1 → sol

Let S = {1, 2, 3, ..., 10}. Let A be the event "the number is even" and B be the event "the number is greater than 6". Find, in set notation, A, B, $A \cup B$, $A \cap B$, $\bar{A}$ and $(A \cup B)^c$, and state $n(\cdot)$ for each.

B12 §4-1 → sol

Find the complement of each event, in words: (a) selecting a vowel from the English alphabet; (b) getting **at least two** heads when three coins are tossed; (c) selecting a student who scored 80 or above.

B13 §4-1 → sol

The probability that a randomly chosen student in a college owns a car is 0.37. Find the probability that the student does **not** own a car.

PART B·2 §4-2 drills — the addition rules

تدريبات القسم 4-2: قواعد الجمع والأحداث المتنافية

For every question, state whether the events are mutually exclusive *before* choosing a formula. That sentence is a method mark.

B14 §4-2 → sol

A and B are mutually exclusive events with P(A) = 0.42 and P(B) = 0.35. Find P(A ∪ B).

B15 §4-2 → sol

P(A) = 0.50, P(B) = 0.40 and P(A ∩ B) = 0.15. Find P(A ∪ B).

B16 §4-2 → sol

P(A) = 0.60, P(B) = 0.30 and P(A ∪ B) = 0.80. Find P(A ∩ B). Are A and B mutually exclusive?

B17 §4-2 → sol

On S = {1, 2, 3, 4, 5, 6}, let A = {1, 3, 5} and B = {2, 4}. Determine whether A and B are mutually exclusive, showing your reasoning.

B18 §4-2 → sol

A single die is rolled. Let A = "an even number" and B = "a number greater than 3". Find P(A ∪ B), and verify your answer by listing A ∪ B.

B19 §4-2 → sol

A box contains 12 red, 8 blue and 5 green marbles. One marble is drawn at random. Find the probability that it is red or green.

B20 §4-2 → sol

Using the box of B19, find the probability that the marble drawn is **not** blue. Comment on your two answers.

B21 §4-2 → sol

In a clinic of 40 patients, 18 have diabetes, 15 have hypertension and 7 have both conditions. For a patient selected at random, find: (a) P(diabetes or hypertension); (b) P(neither condition).

B22 §4-2 → sol

One card is drawn from a standard 52-card deck. Find the probability that it is a king **or** a heart.

B23 §4-2 → sol

Three mutually exclusive events have probabilities 0.18, 0.25 and 0.31. Find the probability that **none** of the three events occurs.

PART B·3

§4-3 drills — multiplication rules and conditional probability

تدريبات القسم 4-3: قواعد الضرب، الاستقلال، والاحتمال الشرطي

Circle the words "with replacement", "without replacement", "given that" and "of those who" before you start. Each one selects a different formula.

B24 §4-3 → sol

A fair coin is tossed twice. Find the probability of getting two heads.

B25 §4-3 → sol

A coin is tossed and a die is rolled. Find the probability of getting a tail on the coin **and** an odd number on the die.

B26 §4-3 → sol

A bag contains 5 red, 3 white and 2 black balls. Two balls are drawn **with replacement**. Find the probability that both are red.

B27 §4-3 → sol

Using the bag of B26, two balls are now drawn **without replacement**. Find the probability that both are red, and compare with B26.

B28 §4-3 → sol

Using the bag of B26 **without replacement**, find the probability of drawing a red ball first and then a black ball.

B29 §4-3 → sol

Approximately 12% of a population is left-handed. Three people are selected at random. Find the probability that all three are left-handed.

B30 §4-3 → sol

Using the data of B29, find the probability that **at least one** of the three people is left-handed.

B31 §4-3 → sol

Given $P(A) = 0.40$ and $P(A \cap B) = 0.12$, find $P(B \mid A)$.

B32 §4-3 → sol

Given $P(B) = 0.25$ and $P(A \mid B) = 0.60$, find $P(A \cap B)$.

B33 §4-3 → sol

$P(A) = 0.50$, $P(B) = 0.30$ and $P(A \cap B) = 0.15$. Determine whether A and B are independent. Are they mutually exclusive?

B34 §4-3 → sol

The results of an examination for 120 students are:

	PASSED	FAILED	TOTAL
Male	48	22	70
Female	38	12	50
Total	**86**	**34**	**120**

For a student selected at random, find: (a) P(female); (b) P(passed | male); (c) P(male | failed); (d) P(female and passed); (e) P(female or passed); (f) are "female" and "passed" independent?

B35 §4-3 → sol

A machine produces parts of which 4% are defective. Two parts are selected independently. Find the probability that **at least one** is defective.

B36 §4-3 → sol

In a college, 70% of students own a laptop. **Of those students**, 40% also own a tablet. Find the probability that a randomly selected student owns both a laptop and a tablet.

B37 §4-3 → sol

Box 1 contains 3 white and 2 black balls. Box 2 contains 1 white and 4 black balls. A fair die is rolled: if it shows 1 or 2, a ball is drawn from Box 1; otherwise a ball is drawn from Box 2. Find the probability that the ball drawn is white.

PART B·4 §4-4 & §4-5 drills — counting rules and probability

تدريبات القسمين 4-4 و 4-5: قواعد العدّ والاحتمال باستخدامها

These two sections carry In-Class Quiz II on their own. For every question, ask "does order matter?" before writing a single symbol.

B38 §4-4 → sol

Evaluate: (a) 6! (b) 0! (c) 8! / 5! (d) (4 + 2)!

B39 §4-4 → sol

Evaluate: (a) ${}_7P_3$ (b) ${}_7C_3$ (c) ${}_9P_2$ (d) ${}_9C_2$. Comment on the relationship between each P and C pair.

B40 §4-4 → sol

A restaurant offers 4 starters, 7 main courses and 3 desserts. How many different three-course meals can be ordered?

B41 §4-4 → sol

A PIN consists of 5 digits chosen from 0–9. How many different PINs are possible if (a) repetition is allowed; (b) repetition is not allowed?

B42 §4-4 → sol

A licence plate consists of 2 letters (from 26) followed by 3 digits (from 0–9), with repetition allowed throughout. How many plates are possible?

B43 §4-4 → sol

In how many different ways can 6 people be seated in a row of 6 chairs?

B44 §4-4 → sol

A club has 10 members. In how many ways can a president, a vice-president and a secretary be chosen? State which counting rule you used and why.

B45 §4-4 → sol

Using the same club of 10 members, in how many ways can a committee of 3 members be chosen? Explain why the answer differs from B44.

B46 §4-4 → sol

How many distinct arrangements can be made from the letters of the word *MISSISSIPPI*?

B47 §4-4 → sol

How many distinct arrangements can be made from the letters of the word *PROBABILITY*?

B48 §4-4 → sol

A class contains 9 girls and 7 boys. A committee of 4 girls and 3 boys is to be formed. In how many ways can this be done?

B49 §4-5 → sol

A box contains 10 good bulbs and 5 defective bulbs. Three bulbs are chosen at random. Find: (a) P(all three are good); (b) P(exactly one is defective); (c) P(at least one is defective).

B50 §4-5 → sol

From a group of 7 men and 5 women, 3 people are selected at random. Find the probability that all three are women.

B51 §4-5 → sol

Using the group of B50, find the probability that exactly 2 men and 1 woman are selected.

B52 §4-5 → sol

A four-digit number is formed at random using the digits 1 to 9, with repetition permitted. Find the probability that all four digits are the same.

B53 §4-5 → sol

In a lottery, 6 numbers are chosen at random from 40, and order does not matter. Find the probability of matching all 6 numbers with a single ticket.

PART C Exam-style structured questions

أسئلة على نمط الامتحان — متعددة الفروع

Each question mixes two or more sections, exactly as a midterm or final question does. Define your events in symbols first; the rest of the solution writes itself.

C1 §4-1 §4-2 → sol

In a sample of 200 people, the blood types recorded were: type O 84, type A 76, type B 28, type AB 12.

1. Set up the frequency distribution and verify the total.
2. Find P(type A).
3. Find P(type B or type AB).
4. Find P(not type O).
5. Find P(neither type A nor type B).

C2 §4-1 §4-3 → sol

Three fair coins are tossed.

1. Write the sample space S in set notation and state n(S).
2. Find P(exactly two heads).
3. Find P(at least one head).
4. Find P(no heads).
5. Verify that your answers to (c) and (d) satisfy the complement rule.

C3 §4-2 §4-3 → sol

A survey of 250 commuters recorded their usual mode of transport:

	CAR	BUS	WALK	TOTAL
Male	78	42	25	145
Female	52	38	15	105
Total	**130**	**80**	**40**	**250**

For a commuter selected at random, find:

1. P(travels by bus).
2. P(female and walks).
3. P(travels by car or is female).
4. P(travels by car | male).
5. P(male | travels by bus).
6. Are "male" and "car" independent? Justify.
7. Are "bus" and "walk" mutually exclusive? Justify.

C4 §4-3 → sol

A box contains 6 blue pens and 5 yellow pens. Three pens are drawn **without replacement**.

1. P(all three are blue).
2. P(the first two are blue and the third is yellow).
3. P(no yellow pen is drawn).
4. P(at least one yellow pen is drawn).

C5 §4-3 → sol

In a factory, machine A produces 60% of the output and machine B produces 40%. Three per cent of machine A's output is defective and five per cent of machine B's output is defective. One item is selected at random.

1. List the four paths of the tree diagram with their probabilities, and verify they sum to 1.
2. Find P(the item is defective).
3. Find P(the item is from machine A **and** is defective).
4. *Stretch:* given that the item is defective, find the probability that it came from machine A.

C6 §4-4 → sol

A student ID consists of 2 letters (A–Z) followed by 4 digits (0–9).

1. How many IDs are possible if repetition is allowed?
2. How many are possible if no letter and no digit may repeat?
3. How many begin with the letter A (repetition otherwise allowed)?

C7 §4-4 → sol

For each of the following, state whether a **permutation** or a **combination** is required, then compute the value.

1. Choosing 3 books to read from 8 available books.
2. Awarding gold, silver and bronze medals among 8 runners.
3. Selecting a 5-player team from 12 players.
4. Arranging 5 different books on a shelf.
5. Choosing a chairperson and a treasurer from 15 members.

C8 §4-5 → sol

A delegation of 5 people is chosen at random from 8 engineers and 6 doctors.

1. Find n(S).
2. P(all five are engineers).
3. P(exactly 3 engineers and 2 doctors).
4. P(at least one doctor).

C9 §4-1 §4-2 §4-3 → sol

In a group of 100 students, 60 study Statistics, 45 study Computing and 25 study both.

1. How many study Statistics only, Computing only, and neither?
2. P(studies Statistics or Computing).
3. P(studies neither).
4. P(studies Computing | studies Statistics).
5. Are "Statistics" and "Computing" independent? Justify.

C10 §4-4 §4-5 → sol

The letters of the word *BANANA* are arranged at random.

1. How many distinct arrangements are there?
2. How many of them begin with the letter B?
3. Find the probability that a randomly chosen arrangement begins with B, and verify your answer by a second method.

PART D — Midterm simulation — 60 minutes, closed book

امتحان محاكٍ للميدتيرم — 60 دقيقة، بدون مراجع

Rules. 60 minutes. Calculator permitted (Casio). No notes. Write every sample space in set notation, name the rule you use before using it, and finish each numerical answer with one sentence of interpretation.

Marking guide. 50 marks total. The real midterm covers Chapters 1–4 in one hour, so treat anything over 45 minutes here as a warning that your Chapter 4 pace is too slow.

D1 §4-1 §4-2 §4-3 → sol

[4 marks] Define each of the following in one sentence: (i) sample space; (ii) simple event; (iii) mutually exclusive events; (iv) independent events.

D2 §4-1 → sol

[4 marks] A survey of 150 households recorded the number of cars owned: 0 cars → 18; 1 car → 62; 2 cars → 49; 3 or more → 21. Find: (a) P(owns exactly 2 cars); (b) P(owns at most 1 car); (c) P(owns at least 2 cars); (d) P(owns fewer than 3 cars).

D3 §4-1 → sol

[4 marks] Let S = {1, 2, ..., 20}. Let A = "a multiple of 3" and B = "a multiple of 4". Write A, B, A ∩ B and A ∪ B in set notation, then find P(A ∪ B) in two different ways.

D4 §4-2 → sol

[3 marks] In a company, 35% of employees speak French, 48% speak German and 12% speak both. Find: (a) P(French or German); (b) P(neither language); (c) P(French only).

D5 §4-3 → sol

[5 marks] A jar contains 7 green and 5 red sweets. Two sweets are taken **without replacement**. Find: (a) P(both green); (b) P(one of each colour); (c) P(both red). Verify that your three answers sum to 1.

D6 §4-3 → sol

[5 marks] Of 300 patients given a diagnostic test:

	DISEASE	NO DISEASE	TOTAL
Test positive	57	21	78
Test negative	3	219	222
Total	**60**	**240**	**300**

Find: (a) P(test positive); (b) P(test positive | disease); (c) P(disease | test positive); (d) are "test positive" and "disease" independent?

D7 §4-3 → sol

[3 marks] Eight per cent of a batch of components are faulty. Four components are chosen independently. Find: (a) P(all four are good); (b) P(at least one is faulty).

D8 §4-4 → sol

[4 marks] Evaluate: (a) ${}_{10}P_4$; (b) ${}_{10}C_4$; (c) $6! / (2! \cdot 3!)$; (d) $0! + 1!$.

D9 §4-4 §4-5 → sol

[5 marks] A password has 6 characters: the first 2 are uppercase letters (26) and the last 4 are digits (0–9), with repetition allowed throughout. Find: (a) the number of possible passwords; (b) how many end in an even digit; (c) the probability that a randomly generated password ends in an even digit.

D10 §4-4 → sol

[3 marks] How many distinct arrangements can be made from the letters of the word *SUCCESS*?

D11 §4-5 → sol

[5 marks] A box contains 9 batteries, 3 of which are dead. Four batteries are selected at random. Find: (a) n(S); (b) P(none is dead); (c) P(exactly 2 are dead); (d) P(at least one is dead).

D12 §4-5 → sol

[5 marks] A committee of 4 is chosen at random from 6 men and 5 women. Find: (a) n(S); (b) P(all four are men); (c) P(exactly 2 men and 2 women); (d) P(at least one woman).

SOLUTIONS Full worked solutions

الحلول الكاملة مع الخطوات

Every answer is shown with the rule named and the substitution written out, because that is how marks are awarded. Where a second method exists, it is given as a check.

Part A — concept and definition questions A1–A25

A1 ← back

(b) an outcome

Definition: an outcome is the result of a *single trial* of a probability experiment. The sample space is the set of *all* outcomes.

A2 ← back

(c) the sample space

Denoted `S`; the number of elements in it is `n(S)`.

A3 ← back

(b) a simple event

An event with two or more outcomes is a **compound** event.

A4 ← back

(c) certain

Probability rule (3). If `P(E) = 0` the event is impossible — rule (2).

A5 ← back

(c) 5/4

Rule (1): `0 ≤ P(E) ≤ 1`, and 5/4 = 1.25 > 1. Use this as a check on every answer you produce.

A6 ← back

(b) empirical

Empirical probability is based on **observation**: `P(E) = f/n`.

A7 ← back

(c) subjective probability

An educated guess or estimate by an expert, employing opinions and inexact information. No sample space counted, no frequency table.

A8 ← back

(b) 36

6 faces × 6 faces = 36 ordered pairs (Example 4-1).

A9 ← back

(c) 8

2 × 2 × 2 = 8: {BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG} (Example 4-3).

A10 ← back

(b) 2, 3, 4

At most → the boundary value 4 is **included**.

A11 ← back

(b) 5, 6

More **than** → the boundary value 4 is **excluded**.

A12 ← back

(c) mutually exclusive

They cannot occur at the same time: $A \cap B = \varnothing$.

A13 ← back

(b) 0

The intersection is the empty set, so its probability is zero. This is exactly why Addition Rule 2 collapses into Rule 1.

A14 ← back

(a) valid for any two events

Rule 2 is the general formula; Rule 1 is its special case when `P(A ∩ B) = 0`. This is why Rule 2 is always the safe choice under exam pressure.

A15 ← back

(c) A, or B, or both

Probability's "or" is always **inclusive** — it is not the everyday English "or", which often implies one but not the other.

A16 ← back

(b) P(B | A) = P(B)

Knowing A occurred does not change the probability of B. Option (a) and (d) both describe **mutually exclusive** events, which is a different idea entirely.

A17 ← back

(b) dependent

The first item is not returned, so both the numerator and the denominator change for the second selection — Multiplication Rule 2 applies.

A18 ← back

(b) P(A ∩ B) / P(A)

The condition goes in the denominator. Putting P(B) underneath is the single most common error in §4-3.

A19 ← back

(b) independent

With two separate coins, one flip does not affect the other. With a **single** coin, "heads" and "tails" are **mutually exclusive** — that is the distinction the lecture notes flag explicitly.

A20 ← back

(b) 1 − P(none)

"At least one" has many ways to happen; its complement "none" has exactly one. Compute the easy side and subtract.

A21 ← back

(b) 1

By definition. It appears inside $_nP_n = n!/0! = n!$ and $_nC_n = n!/(0!\cdot n!) = 1$.

A22 ← back

(b) a permutation

A permutation is an arrangement of objects in a **specific order**.

A23 ← back

(b) $_{10}C_4$

A committee with no titles has no internal order, so order does not matter → combination. $_{10}C_4 = 210$.

A24 ← back

(b) $_nC_r = {}_nP_r \div r!$

A combination is a permutation with the $r!$ internal orderings divided out — which is why C is always ≤ P.

A25 ← back

(b) combinations

Consistency rule. Mixing a permutation numerator with a combination denominator gives an answer too large by a factor of `r!`, and it is the most common wrong answer in §4-5.

Part B·1 — §4-1 drills B1–B13

B1 ← back

S = {1, 2, 3, 4, 5, 6}, n(S) = 6. E = {2, 4, 6}, n(E) = 3.

$$P(E) = \frac{3}{6} = 0.5$$

B2 ← back

E = {5, 6}, n(E) = 2.

$$P(E) = \frac{2}{6} = \frac{1}{3} \approx 0.333$$

Note: "greater than 4" *excludes* 4.

B3 ← back

n(S) = 36. E = {(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)}, n(E) = 6.

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6} \approx 0.167$$

Worth memorising: 7 is the most likely sum, occurring 6 times out of 36.

B4 ← back

"At least 10" means sums of 10, 11 *and* 12 (10 is included).

- Sum 10: (4,6), (5,5), (6,4) → 3
- Sum 11: (5,6), (6,5) → 2
- Sum 12: (6,6) → 1

$$n(E) = 3 + 2 + 1 = 6 \rightarrow P = \frac{6}{36} = \frac{1}{6} \approx 0.167$$

B5 ← back

E = {(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)}, n(E) = 6.

$$P(\text{doubles}) = \frac{6}{36} = \frac{1}{6} \approx \boxed{0.167}$$

B6 ← back

S = {BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG}, n(S) = 8.

"At least two boys" = exactly two boys *or* three boys → E = {BBG, BGB, GBB, BBB}, n(E) = 4.

$$P = \frac{4}{8} = \boxed{0.5}$$

B7 ← back

"No girls" means all three are boys → E = {BBB}, n(E) = 1.

$$P = \frac{1}{8} = \boxed{0.125}$$

B8 ← back

n(S) = 52.

(a) There are 13 hearts:

$$P(\text{heart}) = \frac{13}{52} = \frac{1}{4} = \boxed{0.25}$$

(b) There are 4 kings, one per suit:

$$P(\text{king}) = \frac{4}{52} = \frac{1}{13} \approx \boxed{0.0769}$$

B9 ← back

Check the total: 34 + 28 + 18 = 80 ✔ This is **empirical** probability, `P = f/n`.

1. **(a)** P(tea) = 34/80 = **0.425**
2. **(b)** P(not coffee) = 1 − 28/80 = 52/80 = **0.65**
3. **(c)** Tea and juice are mutually exclusive → Addition Rule 1: P = (34 + 18)/80 = 52/80 = **0.65**

Why (b) and (c) agree: "not coffee" and "tea or juice" describe the same set of people. Spotting that saves a calculation.

B10 ← back

"Neither tea nor coffee" leaves only juice.

$$P = \frac{18}{80} = \boxed{0.225}$$

Or by complement: 1 − (34 + 28)/80 = 1 − 0.775 = 0.225 ✔

B11 ← back

SET	ELEMENTS	N
A (even)	{2, 4, 6, 8, 10}	5
B (> 6)	{7, 8, 9, 10}	4
A ∪ B	{2, 4, 6, 7, 8, 9, 10}	7
A ∩ B	{8, 10}	2
$\bar{A}$	{1, 3, 5, 7, 9}	5
$(A \cup B)^c$	{1, 3, 5}	3

Check: n(A ∪ B) + n($(A \cup B)^c$) = 7 + 3 = 10 = n(S) ✔ Also by the addition rule: 5 + 4 − 2 = 7 ✔

B12 ← back

1. **(a)** `Selecting a consonant.`
2. **(b)** `Getting at most one head` — that is, zero heads or one head.
3. **(c)** `Selecting a student who scored below 80.`

The pattern: the complement of "at least *k*" is "at most *k* − 1". Getting this boundary right is worth a mark on its own.

B13 ← back

Complement rule: `P(Ē) = 1 − P(E)`.

`P(no car) = 1 − 0.37 =` **`0.63`**

Part B·2 — §4-2 drills B14–B23

B14 ← back

Mutually exclusive → **Addition Rule 1**.

$P(A \cup B) = 0.42 + 0.35 =$ **0.77**

B15 ← back

$P(A \cap B) = 0.15 \neq 0$, so the events overlap → **Addition Rule 2**.

$P(A \cup B) = 0.50 + 0.40 - 0.15 =$ **0.75**

B16 ← back

Rearrange Addition Rule 2:

$0.80 = 0.60 + 0.30 - P(A \cap B) \implies P(A \cap B) = 0.90 - 0.80 =$ **0.10**

Since $P(A \cap B) = 0.10 \neq 0$, the events are **not** mutually exclusive.

B17 ← back

$A \cap B = \{1, 3, 5\} \cap \{2, 4\} = \varnothing$

Yes — A and B are mutually exclusive. They share no outcome, so they cannot occur at the same time.

Method mark: writing the intersection and showing it is empty earns more than a bare "yes".

B18 ← back

$A = \{2, 4, 6\}$; $B = \{4, 5, 6\}$; $A \cap B = \{4, 6\}$, so they are **not** mutually exclusive → Rule 2.

$$P(A \cup B) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6} = \frac{2}{3} \approx \textbf{0.667}$$

Verification by listing: $A \cup B = \{2, 4, 5, 6\}$, so $n = 4$ and $P = 4/6$ ✔

B19 ← back

Total = 12 + 8 + 5 = 25. A marble has one colour, so the events are mutually exclusive → Rule 1.

$$P(\text{red} \cup \text{green}) = \frac{12}{25} + \frac{5}{25} = \frac{17}{25} = \boxed{0.68}$$

B20 ← back

$$P(\text{not blue}) = 1 - \frac{8}{25} = \frac{17}{25} = \boxed{0.68}$$

Comment: identical to B19, because "not blue" and "red or green" are the same event. When the categories are exhaustive, the complement is often the faster route.

B21 ← back

7 patients have both, so the events overlap → Rule 2.

1. **(a)**

   $$P(D \cup H) = \frac{18}{40} + \frac{15}{40} - \frac{7}{40} = \frac{26}{40} = \boxed{0.65}$$

2. **(b)**

   $$P(\text{neither}) = 1 - 0.65 = \boxed{0.35}$$

Venn check: diabetes only 11, both 7, hypertension only 8, neither 14. Total 11+7+8+14 = 40 ✔ and 14/40 = 0.35 ✔

B22 ← back

There are 4 kings and 13 hearts, and the **king of hearts is both** — so the events are not mutually exclusive → Rule 2.

$$P(\text{king} \cup \text{heart}) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13} \approx \boxed{0.3077}$$

The classic error is 17/52, obtained by forgetting that one card is counted twice.

B23 ← back

Three mutually exclusive events → extended Addition Rule 1.

$$P(A \cup B \cup C) = 0.18 + 0.25 + 0.31 = 0.74$$

$$P(\text{none}) = 1 - 0.74 = \boxed{0.26}$$

Part B·3 — §4-3 drills B24–B37

B24 ← back

The two tosses are **independent** → Multiplication Rule 1.

$$P(H \cap H) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = \boxed{0.25}$$

B25 ← back

Separate devices → independent. P(tail) = 1/2; P(odd) = 3/6 = 1/2.

$$P = \frac{1}{2} \cdot \frac{1}{2} = \boxed{0.25}$$

Check by sample space: n(S) = 2 × 6 = 12; favourable = T1, T3, T5 → 3/12 = 0.25 ✔

B26 ← back

Total = 5 + 3 + 2 = 10. **With replacement** → independent → Rule 1.

$$P(R \cap R) = \frac{5}{10} \cdot \frac{5}{10} = \frac{25}{100} = \boxed{0.25}$$

B27 ← back

Without replacement → dependent → Rule 2. After one red is removed, 4 reds remain out of 9 balls.

$$P = \frac{5}{10} \cdot \frac{4}{9} = \frac{20}{90} = \frac{2}{9} \approx \boxed{0.2222}$$

Comparison with B26: 0.2222 < 0.25. Removing a red ball makes a second red less likely — which is exactly what "dependent" means.

B28 ← back

Without replacement; the black balls are untouched by removing a red, so 2 blacks remain out of 9.

$$P(R \text{ then } B) = \frac{5}{10} \cdot \frac{2}{9} = \frac{10}{90} = \frac{1}{9} \approx \boxed{0.1111}$$

Note: only the **denominator** drops (10 → 9). The black numerator stays at 2 because no black ball was removed.

B29 ← back

Three people from a large population → independent.

$P(\text{all three}) = (0.12)^3 =$ **0.001728**

B30 ← back

Use the complement: "at least one" ⟹ 1 − P(none).

$P(\text{none left-handed}) = (0.88)^3 = 0.681472$

$P(\text{at least one}) = 1 - 0.681472 =$ **0.3185**

Interpretation: in about 32% of randomly chosen groups of three, at least one person is left-handed.

B31 ← back

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.12}{0.40} = \mathbf{0.30}$$

B32 ← back

Rearranged Multiplication Rule 2: $P(A \cap B) = P(A \mid B) \cdot P(B)$.

$P(A \cap B) = 0.60 \times 0.25 =$ **0.15**

B33 ← back

Independence test: is $P(A \cap B) = P(A) \cdot P(B)$?

$P(A) \cdot P(B) = 0.50 \times 0.30 = 0.15 \quad = \quad P(A \cap B) = 0.15$ ✔

Yes — A and B are independent.

Mutually exclusive? No. $P(A \cap B) = 0.15 \neq 0$, so they *can* occur together. This question exists to prove the two ideas are different: these events are independent *and* not mutually exclusive.

B34 ← back

1. **(a)** P(female) = 50/120 = 5/12 ≈ **0.4167**
2. **(b)** Condition "male" → denominator is the male row total, 70.
   $P(\text{passed} \mid \text{male}) = \frac{48}{70} = \frac{24}{35} \approx$ **0.6857**
3. **(c)** Condition "failed" → denominator is the failed column total, 34.
   $P(\text{male} \mid \text{failed}) = \frac{22}{34} = \frac{11}{17} \approx$ **0.6471**
4. **(d)** A joint "and" → single cell over the grand total.
   $P(\text{female} \cap \text{passed}) = \frac{38}{120} \approx$ **0.3167**
5. **(e)** Addition Rule 2:
   $P(\text{female} \cup \text{passed}) = \frac{50 + 86 - 38}{120} = \frac{98}{120} \approx$ **0.8167**
6. **(f)** P(F)·P(P) = (50/120)(86/120) = 0.4167 × 0.7167 ≈ 0.2986, but P(F ∩ P) = 0.3167. Since 0.3167 ≠ 0.2986 → **not independent (dependent)**.

The denominator rule: "and" keeps the grand total underneath; "given" swaps in the row or column total of the condition. Getting these two the wrong way round is the most frequent table error.

B35 ← back

P(good) = 1 − 0.04 = 0.96; selections are independent.

$P(\text{neither defective}) = (0.96)^2 = 0.9216$

$P(\text{at least one defective}) = 1 - 0.9216 =$ **0.0784**

B36 ← back

"**Of those students**" signals a conditional: P(L) = 0.70 and P(T | L) = 0.40 → Multiplication Rule 2.

$P(L \cap T) = 0.70 \times 0.40 =$ **0.28**

This is Example 4-29 in disguise. The phrase "of those" is the tell every time.

B37 ← back

P(Box 1) = 2/6 = 1/3 · P(Box 2) = 4/6 = 2/3.

PATH	P(BOX)	P(WHITE | BOX)	PRODUCT
Box 1 → white	1/3	3/5	3/15 = 1/5
Box 1 → black	1/3	2/5	2/15
Box 2 → white	2/3	1/5	2/15
Box 2 → black	2/3	4/5	8/15

White is reached by two paths, so **add** them:

$$P(\text{white}) = \frac{3}{15} + \frac{2}{15} = \frac{5}{15} = \frac{1}{3} \approx \boxed{0.3333}$$

Check: 3/15 + 2/15 + 2/15 + 8/15 = 15/15 = 1 ✔ A tree whose branches do not sum to 1 contains an error.

Part B·4 — §4-4 & §4-5 drills B38–B53

B38 ← back

1. **(a)** 6! = 6·5·4·3·2·1 = **720**
2. **(b)** 0! = **1** (by definition)
3. **(c)** 8!/5! = 8·7·6 = **336** — cancel the common 5! instead of expanding both
4. **(d)** (4 + 2)! = 6! = **720** — add inside the bracket first

Casio: for (d), `( 4 + 2 ) SHIFT x⁻¹ (x!) =`

B39 ← back

1. **(a)** ${}_7P_3 = 7!/4! = 7 \cdot 6 \cdot 5 =$ **210**
2. **(b)** ${}_7C_3 = 210/3! = 210/6 =$ **35**
3. **(c)** ${}_9P_2 = 9 \cdot 8 =$ **72**
4. **(d)** ${}_9C_2 = 72/2! =$ **36**

Relationship: ${}_nC_r = {}_nP_r \div r!$ in both pairs. The combination is always the smaller number, because the r! internal orderings have been divided out.

B40 ← back

Fundamental Counting Rule — a sequence of three independent choices.

$4 \times 7 \times 3 =$ **84 meals**

B41 ← back

1. **(a) Repetition allowed** — every slot keeps all 10 digits:
   $10^5 =$ **100,000**
2. **(b) Repetition not allowed** — the pool shrinks by one each time:
   $10 \times 9 \times 8 \times 7 \times 6 =$ **30,240**

This is the lecturer's rule of thumb in action: with repetition the numbers **stay the same** left to right; without repetition they **decrease by 1** each place.

B42 ← back

$26 \times 26 \times 10 \times 10 \times 10 = 26^2 \times 10^3 = 676 \times 1{,}000 =$ **676,000 plates**

B43 ← back

All 6 people arranged in order → a full permutation of 6 objects.

$6! =$ **720 ways**

B44 ← back

Permutation — the three offices are distinct, so who gets which position matters.

$$_{10}P_3 = \frac{10!}{7!} = 10 \times 9 \times 8 = \boxed{720 \text{ ways}}$$

B45 ← back

Combination — a committee with no titles has no internal order.

$$_{10}C_3 = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = \frac{720}{6} = \boxed{120 \text{ ways}}$$

Why it differs from B44: each committee of 3 can be arranged into offices in 3! = 6 ways, so 720 ÷ 6 = 120. The two questions use the same three people; only the question "does order matter?" changes.

B46 ← back

MISSISSIPPI has 11 letters: M = 1, I = 4, S = 4, P = 2. Check: 1 + 4 + 4 + 2 = 11 ✔ → Permutation Rule 2.

$$= \frac{11!}{1! \cdot 4! \cdot 4! \cdot 2!} = \frac{39{,}916{,}800}{1 \cdot 24 \cdot 24 \cdot 2} = \frac{39{,}916{,}800}{1{,}152} = \boxed{34{,}650}$$

B47 ← back

PROBABILITY has 11 letters: B = 2, I = 2, and P, R, O, A, L, T, Y appear once each. Check: 2 + 2 + 7 = 11 ✔

$$= \frac{11!}{2! \cdot 2!} = \frac{39{,}916{,}800}{4} = \boxed{9{,}979{,}200}$$

Always tally the letters first and verify the tally sums to n. Missing a repeated letter is the usual error here.

B48 ← back

Choose girls **and** boys → two combinations multiplied.

$$_9C_4 = \frac{9 \cdot 8 \cdot 7 \cdot 6}{24} = 126 \quad \cdot \quad _7C_3 = \frac{7 \cdot 6 \cdot 5}{6} = 35$$

$$126 \times 35 = \boxed{4{,}410 \text{ ways}}$$

"And" means multiply. Writing 126 + 35 answers a different question.

B49 ← back

10 good + 5 defective = 15 bulbs; choosing 3 with no order → combinations.

$$n(S) = {}_{15}C_3 = \frac{15 \cdot 14 \cdot 13}{6} = 455$$

1. **(a) All good** — all 3 from the 10 good:
   $${}_{10}C_3 = 120 \rightarrow P = \frac{120}{455} = \frac{24}{91} \approx \boxed{0.2637}$$
2. **(b) Exactly one defective** — 1 from 5 defective **and** 2 from 10 good:
   $${}_5C_1 \times {}_{10}C_2 = 5 \times 45 = 225 \rightarrow P = \frac{225}{455} = \frac{45}{91} \approx \boxed{0.4945}$$
3. **(c) At least one defective** — complement of (a):
   $$P = 1 - \frac{24}{91} = \frac{67}{91} \approx \boxed{0.7363}$$

B50 ← back

Total = 12 people; choose 3.

$$n(S) = {}_{12}C_3 = \frac{12 \cdot 11 \cdot 10}{6} = 220 \quad \cdot \quad n(E) = {}_5C_3 = 10$$

$$P = \frac{10}{220} = \frac{1}{22} \approx \boxed{0.0455}$$

B51 ← back

$$n(E) = {}_7C_2 \times {}_5C_1 = 21 \times 5 = 105$$

$$P = \frac{105}{220} = \frac{21}{44} \approx \boxed{0.4773}$$

B52 ← back

Digits 1–9, repetition permitted, four positions.

$$n(S) = 9^4 = 6{,}561 \quad \cdot \quad n(E) = 9 \quad (1111,\ 2222,\ \ldots,\ 9999)$$

$$P = \frac{9}{6{,}561} = \frac{1}{729} \approx \boxed{0.00137}$$

This is Example 4-53 with a different digit pool. Note the pool excludes 0, so it is 9 and not 10.

B53 ← back

Order does not matter → combination.

$$n(S) = {}_{40}C_6 = \frac{40 \cdot 39 \cdot 38 \cdot 37 \cdot 36 \cdot 35}{720} = 3{,}838{,}380$$

$$P(\text{matching all 6}) = \frac{1}{3{,}838{,}380} \approx 2.6 \times 10^{-7}$$

Interpretation: roughly one chance in 3.84 million — about 0.000026%.

Part C — exam-style structured questions C1–C10

C1 ← back

TYPE	FREQUENCY	PROBABILITY
O	84	0.42
A	76	0.38
B	28	0.14
AB	12	0.06
Total	**200**	**1.00** ✔

1. **(a)** 84 + 76 + 28 + 12 = **200** ✔ and the probabilities sum to 1.00, which is probability rule (4).
2. **(b)** P(A) = 76/200 = **0.38**
3. **(c)** Mutually exclusive → Rule 1: P(B ∪ AB) = (28 + 12)/200 = 40/200 = **0.20**
4. **(d)** Complement rule: P(not O) = 1 − 84/200 = 116/200 = **0.58**
5. **(e)** "Neither A nor B" leaves types O and AB: (84 + 12)/200 = 96/200 = **0.48**

Structure to notice: one table, three different tools — addition rule in (c), complement in (d), translated complement in (e). This is precisely how Example 4-13 is built, and how an exam question is built.

C2 ← back

1. **(a)** S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}, **`n(S) = 8`**
2. **(b)** E = {HHT, HTH, THH}, n(E) = 3 → P = 3/8 = **`0.375`**
3. **(c)** Complement of "no heads": P = 1 − 1/8 = 7/8 = **`0.875`**
4. **(d)** E = {TTT} → P = 1/8 = **`0.125`**
5. **(e)** 0.875 + 0.125 = 1.000 ✔ which confirms `P(E) + P(Ē) = 1`.

Alternative for (c): count directly — 7 of the 8 outcomes contain at least one H. The complement route is faster and is the one examiners reward.

C3 ← back

1. **(a)** P(bus) = 80/250 = **`0.32`**
2. **(b)** Joint "and" → cell over grand total: P(female ∩ walk) = 15/250 = **`0.06`**
3. **(c)** Addition Rule 2:

   $$P(\text{car} \cup \text{female}) = \frac{130 + 105 - 52}{250} = \frac{183}{250} = 0.732$$

4. **(d)** Condition "male" → male row total 145:

   $$P(\text{car} \mid \text{male}) = \frac{78}{145} \approx 0.5379$$

5. **(e)** Condition "bus" → bus column total 80:

   $$P(\text{male} \mid \text{bus}) = \frac{42}{80} = 0.525$$

6. **(f)** P(M ∩ C) = 78/250 = 0.312. P(M)·P(C) = (145/250)(130/250) = 0.58 × 0.52 = 0.3016. Since 0.312 ≠ 0.3016 → **`not independent`**.
7. **(g)** A commuter has one usual mode, so Bus ∩ Walk = ∅ → **`yes, mutually exclusive`**.

Parts (f) and (g) are deliberately adjacent. "Mutually exclusive" asks whether the overlap is empty; "independent" asks whether one event changes the other. Answering one with the other's test scores zero.

C4 ← back

6 blue + 5 yellow = 11 pens. Without replacement → dependent → Multiplication Rule 2 applied twice.

1. **(a)**
   $$P(BBB) = \frac{6}{11} \cdot \frac{5}{10} \cdot \frac{4}{9} = \frac{120}{990} = \frac{4}{33} \approx \boxed{0.1212}$$
2. **(b)**
   $$P(B, B, Y) = \frac{6}{11} \cdot \frac{5}{10} \cdot \frac{5}{9} = \frac{150}{990} = \frac{5}{33} \approx \boxed{0.1515}$$

   Note the yellow numerator stays at 5 — no yellow pen has been removed.
3. **(c)** "No yellow" is the same event as "all blue": **4/33 ≈ 0.1212**
4. **(d)** Complement of (c):
   $$P = 1 - \frac{4}{33} = \frac{29}{33} \approx \boxed{0.8788}$$

Check by combinations: P(all blue) = ${}_6C_3/{}_{11}C_3$ = 20/165 = 4/33 ✔ The sequential route and the counting route must agree — use whichever you prefer, but never mix them inside one calculation.

C5 ← back

PATH	P(MACHINE)	P(QUALITY | MACHINE)	PRODUCT
A → defective	0.60	0.03	0.018
A → good	0.60	0.97	0.582
B → defective	0.40	0.05	0.020
B → good	0.40	0.95	0.380
Total			**1.000 ✔**

1. **(a)** The four products above sum to 1.000 ✔
2. **(b)** Defective is reached by two paths, so add:
   $P(D) = 0.018 + 0.020 =$ **0.038**
3. **(c)**
   $P(A \cap D) = 0.60 \times 0.03 =$ **0.018**
4. **(d)** *Stretch* — the conditional formula run in reverse:
   $$P(A \mid D) = \frac{P(A \cap D)}{P(D)} = \frac{0.018}{0.038} = \frac{9}{19} \approx 0.4737$$

Read part (d) carefully. Machine A makes 60% of all output, but only about 47% of the *defective* output, because A's defect rate is lower. Conditioning on "defective" genuinely changes the picture — that is the whole point of conditional probability.

C6 ← back

1. **(a)**
   $26^2 \times 10^4 = 676 \times 10{,}000 =$ **6,760,000**
2. **(b)** Letters without repetition: 26 × 25 = 650. Digits without repetition: 10 × 9 × 8 × 7 = 5,040.
   $650 \times 5{,}040 =$ **3,276,000**
3. **(c)** First letter fixed as A (1 way), the rest free:
   $1 \times 26 \times 10^4 =$ **260,000**

Sanity check on (c): 260,000 / 6,760,000 = 1/26 — exactly the fraction you would expect, since one of 26 letters is fixed.

C7 ← back

	SITUATION	RULE	VALUE
(a)	3 books from 8, order irrelevant	**Combination**	${}_8C_3 = 56$
(b)	Gold, silver, bronze among 8 runners	**Permutation** — the medals are distinct	${}_8P_3 = 336$
(c)	5-player team from 12	**Combination**	${}_{12}C_5 = 792$
(d)	Arranging 5 books on a shelf	**Permutation** (all of them)	$5! = 120$
(e)	Chairperson and treasurer from 15	**Permutation** — distinct offices	${}_{15}P_2 = 210$

The single test: would swapping two chosen items produce a different result? Gold/silver swapped is a different outcome (permutation); two books on a reading list swapped is the same list (combination).

C8 ← back

8 engineers + 6 doctors = 14 people; choose 5, order irrelevant.

1. **(a)**
$$n(S) = {}_{14}C_5 = \frac{14 \cdot 13 \cdot 12 \cdot 11 \cdot 10}{120} = \boxed{2{,}002}$$
2. **(b)**
$${}_8C_5 = 56 \rightarrow P = \frac{56}{2{,}002} = \frac{4}{143} \approx \boxed{0.0280}$$
3. **(c)**
$${}_8C_3 \times {}_6C_2 = 56 \times 15 = 840 \rightarrow P = \frac{840}{2{,}002} = \frac{60}{143} \approx \boxed{0.4196}$$
4. **(d)** Complement of "no doctor", which is (b):
$$P = 1 - \frac{4}{143} = \frac{139}{143} \approx \boxed{0.9720}$$

This is Example 4-51 in a new costume: "all engineers" plays the role of "none defective", and (d) is the complement you have already computed.

C9 ← back

1. **(a)** Statistics only = 60 − 25 = **35**; Computing only = 45 − 25 = **20**; both = 25; neither = 100 − (35 + 25 + 20) = **20**
2. **(b)**
   $$P(S \cup C) = \frac{60 + 45 - 25}{100} = \frac{80}{100} = \mathbf{0.80}$$
3. **(c)**
   $P(\text{neither}) = 1 - 0.80 =$ **0.20**
   ✔ matches the 20 students from (a)
4. **(d)** Condition "Statistics" → denominator 60:
   $$P(C \mid S) = \frac{25}{60} = \frac{5}{12} \approx \mathbf{0.4167}$$
5. **(e)** P(S)·P(C) = 0.60 × 0.45 = 0.27, but P(S ∩ C) = 0.25. Since 0.25 ≠ 0.27 → **not independent**.
   Equivalently: P(C | S) = 0.4167 but P(C) = 0.45 — knowing a student takes Statistics slightly lowers the chance they take Computing.

C10 ← back

BANANA: 6 letters with B = 1, A = 3, N = 2. Check: 1 + 3 + 2 = 6 ✔

1. **(a)** Permutation Rule 2:
   $$\frac{6!}{1! \cdot 3! \cdot 2!} = \frac{720}{12} = \mathbf{60 \text{ arrangements}}$$
2. **(b)** Fix B in the first position; arrange the remaining A, A, A, N, N:
   $$\frac{5!}{3! \cdot 2!} = \frac{120}{12} = \mathbf{10 \text{ arrangements}}$$
3. **(c)**
   $$P(\text{begins with B}) = \frac{10}{60} = \frac{1}{6} \approx \mathbf{0.1667}$$

Second method for (c): exactly one of the six letters is a B, and every position is equally likely to hold it, so P = 1/6 directly ✔ Agreement between two independent routes is the strongest check you can run in an exam.

Part D — midterm simulation solutions D1–D12 · 50 marks

D1 ← back

1. **(i) Sample space** — the set of all possible outcomes of a probability experiment, denoted S. **1 mark**
2. **(ii) Simple event** — an event that consists of exactly one outcome. **1 mark**
3. **(iii) Mutually exclusive events** — two events that cannot occur at the same time, i.e. they have no outcomes in common ($A \cap B = \varnothing$). **1 mark**
4. **(iv) Independent events** — two events such that the occurrence of one does not affect the probability of the other, i.e. $P(B \mid A) = P(B)$. **1 mark**

These four marks are the cheapest on the paper. Learn them as sentences. Do not answer (iii) and (iv) with the same words — the contrast between "cannot occur together" and "does not affect" is what is being tested.

D2 ← back

Check the total: 18 + 62 + 49 + 21 = 150 ✔ Empirical probability, P = f/n.

1. **(a)** Exactly 2 → one class: 49/150 ≈ **0.3267**
2. **(b)** At most 1 → 0 cars and 1 car (1 included): (18 + 62)/150 = 80/150 = 8/15 ≈ **0.5333**
3. **(c)** At least 2 → 2 cars and 3-or-more (2 included): (49 + 21)/150 = 70/150 = 7/15 ≈ **0.4667**
4. **(d)** Fewer than 3 → everything except "3 or more": (18 + 62 + 49)/150 = 129/150 = **0.86**
   Faster: 1 − 21/150 = 0.86 ✔

Check: (b) + (c) = 8/15 + 7/15 = 1 ✔ — "at most 1" and "at least 2" are complements.

D3 ← back

$A = \{3, 6, 9, 12, 15, 18\}$, $n(A) = 6$ · $B = \{4, 8, 12, 16, 20\}$, $n(B) = 5$

$A \cap B = \{12\}$, $n = 1$ · $A \cup B = \{3, 4, 6, 8, 9, 12, 15, 16, 18, 20\}$, $n = 10$

Method 1 — direct count:

$$P(A \cup B) = \frac{10}{20} = \boxed{0.5}$$

Method 2 — Addition Rule 2:

$$P = \frac{6}{20} + \frac{5}{20} - \frac{1}{20} = \frac{10}{20} = \boxed{0.5}$$

✔

12 is the only number in S that is a multiple of both 3 and 4 (the next would be 24, beyond 20). Forgetting to subtract it gives 11/20 — the expected wrong answer.

D4 ← back

1. **(a)** Not mutually exclusive (12% speak both) → Rule 2:
   $0.35 + 0.48 - 0.12 = \boxed{0.71}$
2. **(b)**
   $P(\text{neither}) = 1 - 0.71 = \boxed{0.29}$
3. **(c)**
   $P(\text{French only}) = 0.35 - 0.12 = \boxed{0.23}$

Venn check: French only 0.23 + both 0.12 + German only 0.36 + neither 0.29 = 1.00 ✔

D5 ← back

7 green + 5 red = 12 sweets; without replacement → dependent.

1. **(a)**
$$P(GG) = \frac{7}{12} \cdot \frac{6}{11} = \frac{42}{132} = \frac{7}{22} \approx \boxed{0.3182}$$
2. **(b)** One of each colour happens two ways — green then red, *or* red then green:
$$P = \frac{7}{12} \cdot \frac{5}{11} + \frac{5}{12} \cdot \frac{7}{11} = \frac{35}{132} + \frac{35}{132} = \frac{70}{132} = \frac{35}{66} \approx \boxed{0.5303}$$
3. **(c)**
$$P(RR) = \frac{5}{12} \cdot \frac{4}{11} = \frac{20}{132} = \frac{5}{33} \approx \boxed{0.1515}$$

Verification: 42/132 + 70/132 + 20/132 = 132/132 = 1 ✔ The three cases are exhaustive, so they must sum to 1. **Part (b) is where marks are lost** — forgetting the second order halves the answer.

D6 ← back

1. **(a)** P(test +) = 78/300 = **0.26**
2. **(b)** Condition "disease" → column total 60:
$$P(+ \mid D) = \frac{57}{60} = \boxed{0.95}$$
3. **(c)** Condition "test +" → row total 78:
$$P(D \mid +) = \frac{57}{78} = \frac{19}{26} \approx \boxed{0.7308}$$
4. **(d)** $P(+ \cap D) = 57/300 = 0.19$; $P(+) \cdot P(D) = 0.26 \times 0.20 = 0.052$. Since $0.19 \neq 0.052$ → **not independent (strongly dependent)**.

Why (b) ≠ (c): the test catches 95% of people who have the disease, but only 73% of positive results are true cases — because far more people are healthy than diseased. Reversing a conditional is not allowed; $P(A|B) \neq P(B|A)$, and this table shows exactly how far apart they can be.

D7 ← back

1. **(a)** P(good) = 0.92 each, independent:
   $(0.92)^4$ = **0.7164**
2. **(b)**
   P(at least one faulty) = 1 − 0.7164 = **0.2836**

Interpretation: although only 8% of individual components are faulty, nearly 28% of batches of four contain at least one fault.

D8 ← back

1. **(a)** ${}_{10}P_4 = 10 \times 9 \times 8 \times 7 =$ **5,040**
2. **(b)** ${}_{10}C_4 = 5{,}040 / 4! = 5{,}040 / 24 =$ **210**
3. **(c)** $6!/(2!\cdot 3!) = 720/(2 \times 6) = 720/12 =$ **60**
4. **(d)** $0! + 1! = 1 + 1 =$ **2**

D9 ← back

1. **(a)**
   $26^2 \times 10^4 = 676 \times 10{,}000 =$ **6,760,000**
2. **(b)** The even digits are 0, 2, 4, 6, 8 → 5 choices for the final character:
   $26^2 \times 10^3 \times 5 = 676 \times 1{,}000 \times 5 =$ **3,380,000**
3. **(c)**
   $$P = \frac{3{,}380{,}000}{6{,}760{,}000} =$$ **0.5**

Sanity check on (c): 5 of the 10 digits are even and the last character is chosen freely, so 0.5 is exactly right. If your counting gives anything else, the counting is wrong.

D10 ← back

SUCCESS has 7 letters: S = 3, C = 2, U = 1, E = 1. Check: 3 + 2 + 1 + 1 = 7 ✔

$$= \frac{7!}{3! \cdot 2! \cdot 1! \cdot 1!} = \frac{5{,}040}{6 \times 2} = \frac{5{,}040}{12} =$$ **420 arrangements**

D11 ← back

3 dead + 6 live = 9 batteries; choose 4, order irrelevant.

1. **(a)**
$$n(S) = {}_9C_4 = \frac{9\cdot 8\cdot 7\cdot 6}{24} = \boxed{126}$$
2. **(b)** All 4 from the 6 live:
$${}_6C_4 = 15 \to P = \frac{15}{126} = \frac{5}{42} \approx \boxed{0.1190}$$
3. **(c)** 2 from the 3 dead **and** 2 from the 6 live:
$${}_3C_2 \times {}_6C_2 = 3 \times 15 = 45 \to P = \frac{45}{126} = \frac{5}{14} \approx \boxed{0.3571}$$
4. **(d)** Complement of (b):
$$P = 1 - \frac{5}{42} = \frac{37}{42} \approx \boxed{0.8810}$$

D12 ← back

6 men + 5 women = 11 people; choose 4.

1. **(a)**
$$n(S) = {}_{11}C_4 = \frac{11\cdot 10\cdot 9\cdot 8}{24} = \boxed{330}$$
2. **(b)**
$${}_6C_4 = 15 \to P = \frac{15}{330} = \frac{1}{22} \approx \boxed{0.0455}$$
3. **(c)**
$${}_6C_2 \times {}_5C_2 = 15 \times 10 = 150 \to P = \frac{150}{330} = \frac{5}{11} \approx \boxed{0.4545}$$
4. **(d)** Complement of "all men":
$$P = 1 - \frac{1}{22} = \frac{21}{22} \approx \boxed{0.9545}$$

Pattern recognition: D11 and D12 are the same question with different nouns, and both are Example 4-51 restated. Once you see that every §4-5 question has the shape *count S, count E, divide, take a complement for "at least"*, the section stops being difficult.

TRACKER

Score tracker

جدول متابعة الدرجات — أعد المحاولة بعد ثلاثة أيام

Print this page or copy the table. Record attempt 1 today and attempt 2 three days later — the gap is what makes the recall stick. Any row scoring below 80% on attempt 2 means re-reading that section of the summary document before anything else.

PART	SECTION	OUT OF	ATTEMPT 1	ATTEMPT 2	WEAK SPOTS TO REVISIT
A	Concepts & definitions (all sections)	25			
B·1	§4-1 Sample spaces & probability	13			
B·2	§4-2 Addition rules	10			
B·3	§4-3 Multiplication & conditional	14			
B·4	§4-4 & §4-5 Counting rules	16			
C	Exam-style structured	10			
D	Midterm simulation (50 marks)	12			
Total		**100**			

READING YOUR SCORE

90–100: A-standard on this chapter. Move to Chapter 5 and return here once, a week before the midterm.
75–89: solid, but the lost marks are concentrated somewhere — find the pattern in the "weak spots" column rather than re-doing everything.
Below 75: do not grind more questions yet. Re-read the matching sections of the summary, rewrite the formula sheet from memory, then come back in two days.

قراءة النتيجة

90–100: مستوى A في هذا الفصل — انتقل إلى الفصل الخامس وعُد لهذا البنك مرة واحدة قبل الميدتيرم بأسبوع.
75–89: جيد، لكن الدرجات الضائعة متركّزة في موضع واحد — ابحث عن النمط في عمود "نقاط الضعف" بدل إعادة كل شيء.
أقل من 75: لا تحلّ أسئلة أكثر الآن. أعد قراءة الأقسام المعنية في ملف الملخص، ثم اكتب ورقة القوانين من ذاكرتك، وعُد بعد يومين.

STAT 2101 — Introduction to Statistics · Chapter 4 Question Bank. One hundred questions with full worked solutions, built against the SQU College of Science course outline (OQF Level 5, Fall 2026) and Dr. Ahmed Al-Hadhrami's Chapter 4 lecture slides. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017).

Every question is tagged with the section it tests, so a wrong answer points straight at the lesson to revisit. This bank is the companion to the Chapter 4 master summary; the two are designed to be used together.

Note on coverage: the outline also assigns textbook exercises from page 247, numbers 6, 8, 16 and 40, plus the Moodle exercise sets for Weeks 5 and 6. Those problems are not reproduced here — do them from the textbook itself, because they are the ones your tutor holds.